TEXORA · LECTURE SHEET

Special Fabric Types — A Survey

Fourteen fabrics that ordinary plain, twill and jersey production does not cover: for each, its definition and structure, how it is produced, the properties the structure buys, and where it is used.

FM-4103 · SPECIAL FABRIC PRODUCTION · TOPIC 1 · FABRIC MANUFACTURING SPECIALIZATION

1. What makes a fabric "special"?

The fabrics of this course are special in one of four ways: the **structure is three-dimensional or piled** (velvet, terry, carpet, spacer), the **product is narrow or shaped** rather than a sheet (tape, belt, label, braid), the fabric is a **composite** of layers or materials (laminate, coated, bonded), or the geometry departs from the rectangular warp–weft grid (tri-axial). Each family below is treated in the same order: definition and structure, production, properties, end-uses.

2. Laminated fabrics

Definition/structure: two or more pre-formed layers permanently joined face-to-face — classically a woven or knitted face, a functional middle (a waterproof-breathable membrane film, or a foam layer), and often a knitted lining, acting as one fabric. **Production:** adhesive lamination (gravure-applied hot-melt or reactive polyurethane dots/lines), flame lamination (a foam surface melted to tack the layers), or direct membrane transfer; the adhesive pattern must bond without sealing the pores that give breathability. **Properties:** each layer keeps its own gift — the face gives abrasion resistance and appearance, the membrane waterproofness with moisture-vapour transmission, foam gives insulation and cushioning; the composite is stiffer than its parts and can delaminate if the bond fails. **End-uses:** rainwear and outdoor shells, footwear uppers, medical barrier fabrics, automotive interiors, mattress protectors.

3. Tape and webbing

Definition/structure: narrow woven fabric, typically up to ~30 cm wide but usually far narrower, woven with finished selvedges on both edges; structures are plain, twill or satin in miniature, often with a thicker "catch" edge. **Production:** needle looms (shuttleless, weaving several tapes side by side across the width), with heavy weft for body. **Properties:** high tensile strength for width, low edge

fraying, compact handle. **End-uses:** bag handles and straps, seat-belt webbing (a critical, standardised product), tie tapes, zipper tapes, elastic tapes, hook-and-loop backing.

4. Belts (industrial belting fabrics)

Definition/structure: heavy, thick narrow fabrics — or wide fabrics cut to width — used as conveyor and transmission belting bases: multi-layer woven constructions, often with folded/plied warp and coarse weft, giving a near-solid pad. **Production:** heavy looms with strong beat-up; commonly impregnated or coated with rubber/PVC after weaving. **Properties:** very high tensile strength and impact absorption, low extension (dimensional stability under load is the design target), abrasion resistance. **End-uses:** conveyor belts for mines and agriculture, power-transmission belting, elevator belts, timing-belt reinforcement.

5. Label fabrics

Definition/structure: narrow satin- or twill-faced fabrics carrying woven-in text and logos — the brand label inside a garment. The ground is usually polyester satin (smooth, lustrous, takes fine detail); the lettering is formed by contrasting weft or warp floats. **Production:** narrow jacquard or dobby needle looms, often weaving many labels across the width, then slit and folded (end-fold, centre-fold, mitre-fold) and cut by heat or ultrasonics to seal edges. **Properties:** permanence — the information cannot wash off as printing can; softness against skin is a design requirement (hence satin faces and sealed, non-scratch edges). **End-uses:** brand, size and care labels; woven badges and patches.

6. Braids

Definition/structure: a set of yarns (commonly 8–144 carriers) interlaced diagonally: each yarn travels helically, passing alternately over and under the yarns travelling the opposite way, at a **braid angle** to the axis. Flat braids are ribbons; tubular braids are hollow sleeves. **Production:** maypole braiding machines — carriers follow interlocking serpentine tracks around the machine, exchanging positions at the braid point; the product is drawn off at a rate that fixes the braid angle. **Properties:** the trellis geometry of Topic 4 (FM-4203) in a tube: braids expand and contract radially, conform to shapes, resist unraveling when cut (ends lock), and carry load in two diagonal directions. **End-uses:** shoelaces and cords, cable sleeving and insulation, surgical sutures and stents (medical braids), composite preforms, wicks, packings.

one yarn set highlighted: it crosses over and under the opposing set in turn



Figure 1 — Flat braid geometry (schematic): two yarn families travel diagonally in opposite senses, interlacing 1/1; the braid angle to the axis is set by carrier speed against take-off rate, and with it the balance of coverage and flexibility.

7. Lappet and swivel fabrics

Both are **figured fabrics woven with extra patterning threads on a plain ground** — the ancestors of embroidery-effect weaving. **Lappet:** extra warp threads, carried on a shifting needle frame above the cloth, are laid into the shed and bound by the ground weft, forming continuous warp-direction patterns (spots, sprigs, stripes of figure) that run the length of the cloth; the patterning threads travel sideways between repeats under needle-frame control. **Swivel:** extra *weft* threads are inserted by small shuttles (swivel shuttles) that weave a figure only where it is needed, interlacing a few ends and returning — producing isolated spots of extra-weft colour, each with its own tiny shuttle and bobbin, at a heavy cost in loom speed. **Properties:** genuine woven-in figure with embroidery-like effect, pattern threads float on the back between motifs (lappet) or are localised (swivel). **End-uses:** historically dress goods, curtains and handkerchief borders; today largely displaced by embroidery and jacquard, and studied as the mechanism ancestors of both.

8. Velvets and velveteen

Definition/structure: warp-pile fabrics — velvet — have a dense erect pile of cut warp ends standing over a plain or twill ground; velveteen is the weft-pile counterpart (an all-cotton, shorter-piled cloth; see also corduroy in TE-3205). **Production:** velvet is woven face-to-face as a double cloth — two grounds joined by the pile warp — and slit apart on the loom or in finishing so each cloth keeps half the pile height; single-cloth velvet uses fine wires withdrawn to cut the loops. Pile density, height and the W or V anchoring of each tuft decide quality. **Properties:** deep lustre from light reflected off the aligned cut ends, rich colour depth, warm handle; the pile crushes and marks with pressure (a real care limitation) and shading differences between pieces (pile direction) are a classic garment-making trap. **End-uses:** evening and formal wear, upholstery, curtains, theatre drapes, jewellery display.

9. Carpets (woven and tufted)

Definition/structure: floor coverings built for thickness and resilience: a pile surface (the wear layer), a primary structure that holds it, and usually a backing that stabilises it. **Production:** (i) *Woven* — Wilton (pile wires, jacquard-patterned, the premium route) and Axminster (tufts inserted row by row, unlimited colour), both slow and costly; (ii) *Tufted* — needles punch pile yarn through a woven or nonwoven primary backing on a wide multi-needle machine, loops are cut or left, and a latex coating plus secondary backing locks the tufts — the dominant world route by volume; (iii) needle-punched and bonded carpets for contract use. **Properties:** underfoot resilience and insulation, sound absorption, pattern fastness decided by yarn dyeing (solution-dyed synthetics dominate contract work); wear is graded by pile density and tuft bind, the subject of TE-3201's carpet testing. **End-uses:** domestic and contract flooring, automotive carpet, mats and rugs.

10. Bonding and lamination of fabric layers (bonded fabrics)

Distinct from the film laminates of Section 2: **bonded fabrics** join fabric to fabric (or fabric to foam/nonwoven) by an adhesive web, powder, or foam-flame process, chiefly to give a cheap or unstable face fabric the body of a backing — e.g. a lace or loose knit bonded to a tricot for garment use, or foam-backed upholstery (the foam route merges into flame lamination). Properties are the sum of the layers minus flexibility; the recurring failure is bond peel, tested as bond strength. End-uses: apparel interlining-backed cloths, upholstery, lampshades, automotive headliners.

11. Terry towel

Definition/structure: a woven fabric with uncut loops of pile warp standing on both faces of a plain ground — the loops are the product. **Production:** a special reed motion (loose pick beating, "Turkish towelling" motion): groups of picks are beaten loosely at a distance from the cloth fell, then driven home together, letting the slack pile warp — woven under low tension from its own beam — buckle into loops between the tight ground warp; typical loop formation uses a 3-pick (or 4-pick) grouping. **Properties:** absorbency from the enormous surface area and capillary volume of the loops; loop height, pile ratio and yarn twist (low twist absorbs, higher twist lasts) are the design variables; a worked absorbency comparison belongs to TE-3201's testing. **End-uses:** towels, bathrobes, bath mats, cleaning cloths.

12. Coated fabrics

Definition/structure: a textile base (woven, knitted or nonwoven) carrying a continuous polymer layer — PVC, polyurethane, silicone, rubber or acrylic — applied as paste, foam or film. **Production:** direct coating (knife-over-roll or knife-over-air spreading of polymer paste, then gelling/curing), transfer coating (film formed on release paper, then transferred — the route to artificial leather), foam coating,

and dip coating; coat weight and penetration are controlled so the layer anchors without striking through. **Properties:** the polymer gives waterproofness, chemical resistance, airtightness or a leather-like surface; the textile gives tear strength the film alone lacks. Breathable PU coatings carry vapour through hydrophilic or microporous structures. **End-uses:** artificial leather (upholstery, footwear, bags), tarpaulins, rainwear, inflatable structures, conveyor surfaces, medical and protective clothing.

13. Filter fabrics

Definition/structure: engineered textiles whose job is controlled permeability: they must pass a fluid and stop particles of a stated size, then release the collected cake and survive cleaning. Woven monofilament cloths give precise, sieve-like pores; needlefelt nonwovens give depth filtration with tortuous paths. **Production:** precision weaving (often monofilament polyester/polypropylene), calendering and singeing to set pore size and surface release; needlefelts are needled, singed and sometimes membrane-laminated (merging with Section 2). **Properties (the specification set):** air/fluid permeability, pore size, particle retention, cake release, chemical and temperature resistance, dimensional stability under pressure. **End-uses:** dust collection (baghouses), liquid filtration in chemical and food plants, water treatment, vacuum and belt filters, HEPA media (as supported nonwoven).

14. Tri-axial fabrics

Definition/structure: three yarn systems interlaced at angles — typically warp at 0° and two weft systems at ±60° — so the cloth is reinforced in three directions instead of two, woven on a special loom whose shed geometry rotates the insertion. **Properties:** near-isotropic in-plane behaviour: no weak bias direction, outstanding shear resistance (the trellis of FM-4203 is locked by the third family), high tear resistance and dimensional stability; the cost is a complex, slow machine and limited widths. **End-uses:** composite reinforcement, sails and membranes, conveyor and structural fabrics, aerospace and defence textiles — wherever a biaxial cloth's 45° weakness is unacceptable.

Glossary

Lamination — permanent face-to-face joining of pre-formed layers. **Webbing** — heavy narrow woven tape for load-bearing straps. **Braid angle** — angle of braid yarns to the product axis. **Face-to-face weaving** — double cloth slit apart to yield two pile fabrics. **Tufting** — punching pile yarn through a pre-formed backing. **Pile ratio** — pile-warp length consumed per unit cloth length in terry. **Tri-axial** — three yarn systems at mutual angles, typically 0/±60°.

Key points

• "Special" = pile/3D structure, narrow/shaped product, composite layers, or non-rectangular geometry.
• Laminates and coated fabrics divide the work: textile for strength, polymer/membrane for barrier — and the bond is the weak point in both.
• Braids, lappet and swivel show pattern and shape built into the forming machine, not added afterwards.
• Filter and tri-axial fabrics are specified by engineering numbers (pore size, permeability, isotropy), the clearest cases of fabric as a designed material.

Practice

1. A customer needs a waterproof, breathable shell fabric for mountaineering jackets. Choose between a coated fabric and a membrane laminate and justify the choice in two structure-based points.
2. Why is a braid, rather than a woven tape, chosen as a cable sleeve that must be pushed over connectors wider than the cable?
3. State the structural difference between velvet and velveteen, and between a Wilton and a tufted carpet.
4. Give the mechanism by which terry loops form, naming the loom motion involved.

Answer key

1. **Membrane laminate.** (i) The membrane is a continuous engineered film laminated with a patterned adhesive, so waterproofness and moisture-vapour transmission are controlled independently of the face fabric's pores; a direct coating tends to seal the cloth as a single layer with lower breathability and a rubbery handle. (ii) The laminate's face fabric protects the fragile membrane from abrasion in use, and a lining protects it inside — the three-layer division of labour suits a garment that is flexed and abraded daily.
2. The braid's diagonal trellis lets the sleeve **expand radially** when pushed (yarns rotate, diameter grows, length shortens — the concertina effect) and then contract to grip the cable; a woven tape's fixed rectangular grid has no such radial compliance and would jam at the connector.
3. Velvet's pile is **warp** pile (cut pile warp over a ground, usually woven face-to-face); velveteen's pile is **weft** pile (extra weft floats cut in finishing). A Wilton carpet is **woven** — pile yarns are woven in over wires as the cloth forms; a tufted carpet's pile is **punched into a pre-woven backing** and locked by a latex coat — different forming principle, speed and cost structure.
4. Loops form by the **slack (loose-reed / terry) beat-up motion**: successive picks (groups of 3–4) are first beaten only part-way toward the fell, leaving the low-tension pile warp slack, then the whole group is driven home together; the pile warp, fed longer than the ground for the same cloth length, has nowhere to go but out of the cloth plane, buckling into loops on both faces.